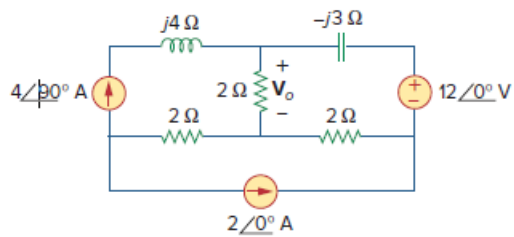


Problem

Compute V_o in the circuit of Fig. 10.81 using mesh analysis.

Figure 10.81:



Step-by-step solution

Step 1 of 3

Refer to Figure 10.81 in the textbook.

Consider I_1 is the current passing in loop 1 and $I_1 = 4\angle 90^\circ$ A.

Consider I_3 is the current passing in loop 3 and $I_3 = -2\angle 0^\circ$ A.

Apply mesh analysis in loop 2.

$$4I_2 - j3I_2 - 2I_1 - 2I_3 + 12 = 0$$

$$(4 - j3)I_2 - 2I_1 - 2I_3 + 12 = 0 \quad \dots (1)$$

Step 2 of 3

Replace $4\angle 90^\circ$ A for I_1 , $-2\angle 0^\circ$ A for I_3 in equation 1.

$$(4 - j3)I_2 - 2(4\angle 90^\circ) - 2(-2) + 12 = 0$$

$$(4 - j3)I_2 - 2(j4) - 2(-2) + 12 = 0$$

$$(4 - j3)I_2 = -12 - 4 + j8$$

$$I_2 = \left(\frac{-16 + j8}{(4 - j3)} \right)$$

$$I_2 = -3.52 - j0.64$$

Step 3 of 3

Determine the value of V_o .

Consider the equation for V_o .

$$V_o = 2(I_1 - I_2) \dots\dots (2)$$

Replace $j4$ for I_1 , and $-3.52 - j0.64$ for I_2 in equation 2.

$$V_o = 2[j4 - (-3.52 - j0.64)]$$

$$= 2[3.52 + j4.64]$$

$$= 7.04 + j9.28$$

$$= 11.64 \angle 52.81^\circ \text{ V}$$

Therefore, the value V_o is $11.64 \angle 52.81^\circ \text{ V}$.